CHAPTER

# Sparse Reconstruction

8

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## 8.1 Introduction

The quest for fast imaging speed started early in the history of MRI. Fast MRI acquisition helps shorten exam times and thus reduces patient discomfort during MRI exams. Moreover, increased imaging speed can be leveraged to improve spatiotemporal resolution and/or volumetric coverage, all of which play an important role in routine clinical applications. Parallel imaging (or parallel MRI, see Chapter 6), introduced in the 1990s, has proven to be one of the most disruptive fast MRI techniques, serving to robustly generate an MR image from undersampled measurements using coil arrays [1–3]. Parallel imaging was later extended via many different variants [4–7], and it is now available in most MRI scanners for routine clinical use. However, the acceleration capability and reconstruction performance of parallel imaging are limited by the design of coil arrays and the number of coil elements, and it is ultimately restricted by fundamental electrodynamic principles [8,9]. In routine clinical applications, parallel imaging typically allows for a moderate acceleration rate, and excessive acceleration often results in noise amplification that degrades image quality [10].

The rise of compressed sensing in the 2000s [11,12] has generated great excitement, and its application to MRI (known as compressed sensing MRI or sparse MRI) has introduced a new and powerful approach to improve imaging speed by exploiting image sparsity/compressibility [13]. Here, image sparsity/compressibility can be understood in a way that most images can be represented with much less data than the number of pixels, due to correlations between pixels [14]. Fig. 8.1 shows a specific example, where a brain MR image can be represented by only 10% of its largest wavelet coefficients without perceptual loss of important information. This implies a so-called transform coding framework that plays an essential role behind the JPEG, JPEG2000, and MPEG image/video compression standards and is widely used in daily life. The ability to compress images so effectively raises a question: Instead of spending so much effort to acquire an image and then discarding most of its coefficients for compression, why not directly acquire the image in a compressed form? In other words, knowing that images are compressible, perhaps we can build the compression process directly into the sampling/sensing step so that one only acquires the information that is needed to represent the image (e.g., acquisition based on the information rate instead of the Nyquist–Shannon rate). This hypothesis was proved by Candes et al., and Donoho (2006), who established a theory that is well known as compressed sensing or compressive sampling [11,12]. Soon after that, Lustig et al. demonstrated that an MR image, under certain conditions, can be successfully recovered from a number of measurements that are far below the Nyquist–Shannon limit [13,14]. About the same time, Block et al. also demonstrated that an MR image can be reconstructed from undersampled multicoil radial measurements by exploiting im-

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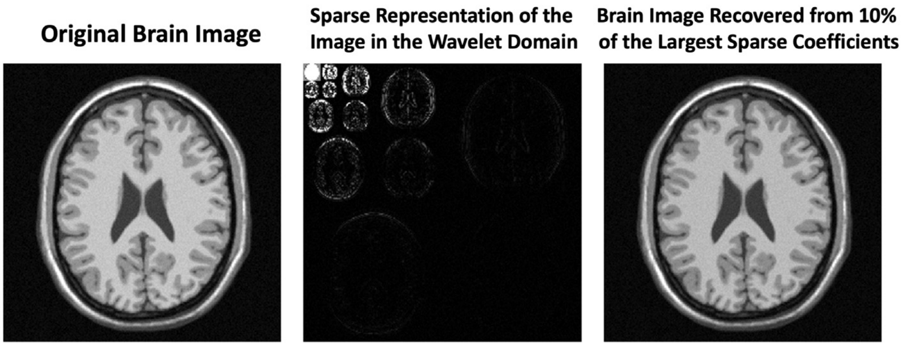



**FIGURE 8.1**

Demonstration of the compressibility of a brain MR image. Although the image is not sparse by itself, it has a sparse representation in the wavelet domain after applying the wavelet transform to the image (e.g., most of coefficients have a value close to zero in the wavelet domain). With an inverse wavelet transform using only 10% of the largest wavelet coefficients, the image can be recovered without perceptual loss of information. This example demonstrates the compressibility of a brain MR image.

age sparsity [15]. Since then, there has been an explosive growth of various sparsity-based fast imaging techniques [16–18], and various studies have also shown that appropriate combinations of compressed sensing MRI with parallel imaging can enable higher acceleration and improved reconstruction quality compared to either of them alone [19–23]. These advances have all synergistically led to remarkably increased imaging speed with previously unachieveable imaging performance [18,24].

Sparse image reconstruction has a long history and goes far beyond standard compressed sensing MRI. Although the role of image sparsity in image reconstruction was not formally established until the advent of compressed sensing, many early constrained reconstruction methods, such as the keyhole [25], TRICKS/HYPR [26,27], and k-t BLAST [6] methods (see Chapter 5), took advantage of spatiotemporal correlations in dynamic MR images, which represent a different notion of image sparsity. In the meantime, most post-compressed sensing image-reconstruction techniques, such as low-rank or dictionary learning-based reconstruction methods, which will be described in Chapters 9–10, also rely on image sparsity. Artificial intelligence-enabled image reconstruction, as the most recent trend in rapid MRI [28–30] that will be reviewed in Chapter 11, also finds its root in the use of image sparsity/correlations and shares certain similarities with the compressed sensing framework. These advances all indicate that we have entered an era of sparsity for rapid MRI, and understanding sparse reconstruction will help pave a path to further develop more powerful and more advanced image reconstruction techniques to tackle yet more clinical problems.

This chapter will focus on compressed sensing and its applications in rapid MRI. It is organized as follows: Section 8.2 briefly reviews the compressed sensing theory; Section 8.3 presents the translation of compressed sensing to MRI and the specific requirements for data acquisition and image reconstruction; Section 8.4 describes representative frameworks to combine compressed sensing MRI with parallel imaging and shows how they can synergistically result in improved reconstruction quality; Section 8.5 describes representative clinical applications of compressed sensing MRI; Sections 8.6 and 8.7 then conclude the chapter with discussion of the existing challenges of compressed sensing

MRI and potential solutions, along with a summary of the chapter. Finally, a MATLAB® tutorial that demonstrates compressed sensing reconstruction for an undersampled brain MRI data is presented in Section 8.8. With knowledge of sparsity and incoherence, the readers will see how they can be further used to develop the more advanced sparse reconstruction methods that are covered in the following chapters.

## 8.2 Compressed sensing theory: a brief overview

Compressed sensing involves acquiring a sparse/compressible signal in an efficient manner using an incoherent measurement basis and then recovering the signal without loss of important information [31]. Here, efficient acquisition is achieved by sampling the signal at a rate that is far below the Nyquist–Shannon requirement. As a result, there are three immediate questions related to compressed sensing: (i) why sparsity is needed and useful; (ii) what incoherent sampling is and why we need it; and (iii) how the sparse signal is recovered. This section briefly reviews the answers to these questions and helps readers better understand the principles behind compressed sensing MRI.

### 8.2.1 Sparsity and incoherence: a first look

We first define a one-dimensional (1D) sensing problem that is typically implemented for signal acquisition and is related to that described in Chapter 2. The sampling of a signal vector $\mathbf{x}$ (size $= N \times 1$) can be interpreted as a projection of the signal onto different sampling waveforms $\mathbf{e}_i$ with the same size:

$$s_i = \langle \mathbf{x}, \mathbf{e}_i \rangle \quad i = 1, 2, 3, ..., M. \tag{8.1}$$

Here, $M$ denotes the total number of measurements and $s_i$ denotes the $i^{th}$ measurement sampled with the $i^{th}$ waveform. In matrix notation, this becomes

$$\mathbf{s} = \mathbf{E}\mathbf{x}, \tag{8.2}$$

where $\mathbf{E}$ denotes a measurement matrix consisting of a total of $M$ sampling waveforms (size $= M \times N$), which is also referred to as an encoding operator, and $\mathbf{s}$ denotes the measurements (size $= M \times 1$). Connecting this to MRI reconstruction, $\mathbf{s}$ can be treated as the k-space measurements that are concatenated into a vector; $\mathbf{x}$ can be treated as the digitalized MR image to be acquired in the form of a vector; and $\mathbf{E}$ can treated as a transform that operates between image and k-space without loss of generality. As seen in Chapter 2, when $M = N$ and the encoding operator $\mathbf{E}$ has a full rank, Eq. (8.2) is a well-defined linear equation with a unique solution (Fig. 8.2a). However, when the number of measurements is smaller than the length of the signal ($M < N$), Eq. (8.2) is an under-determined or undersampled linear equation with an infinite number of possible solutions (Fig. 8.2b). Here comes the role of sparsity: if $\mathbf{x}$ is a $K$-sparse signal (i.e., $\mathbf{x}$ has at most $K$ nonzero coefficients and $K \ll N$) and the locations of the nonzero components are also known (Fig. 8.2c), the signal $\mathbf{x}$ then becomes $\mathbf{x}_K$ with reduced dimensionality (size $= K \times 1$). In a similar way, $\mathbf{E}$ can be reduced to $\mathbf{E}_K$ with size $M \times K$ by keeping only the $K$ columns corresponding to the nonzero locations in $\mathbf{x}$ (see Fig. 8.2c). With the knowledge of signal sparsity and sparse locations, Eq. (8.2) becomes Eq. (8.3), and, assuming that $\mathbf{E}_K$ has a full

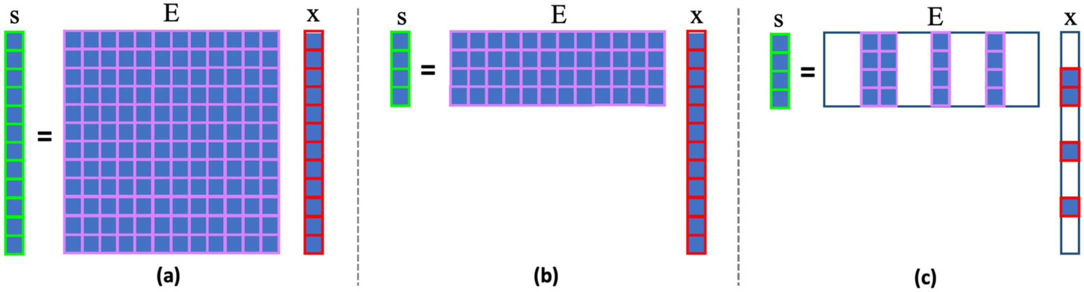



**FIGURE 8.2**

**(a)** A linear equation is well-determined when the number of measurements ($\mathbf{s}$, size $= 12 \times 1$) equals to the length of the signal ($\mathbf{x}$, size $= 12 \times 1$) to be sampled, which ensures a unique solution if $\mathbf{E}$ has a full rank. **(b)** The linear equation becomes under-determined when the number of measurements ($\mathbf{s}$, size $= 4 \times 1$) is smaller than the length of the signal ($\mathbf{x}$, size $= 12 \times 1$) to be sampled, which results in more than one possible solution. **(c)** if $\mathbf{x}$ is a $K$-sparse with $K \ll 12$ (e.g., $\mathbf{x}$ has at most $K = 4$ nonzero coefficients in this example) and the locations of the nonzero components are known, the signal $\mathbf{x}$ then becomes $\mathbf{x}_K$ with reduced dimensionality (size $= 4 \times 1$), and $\mathbf{E}$ also reduces to $\mathbf{E}_K$ with a reduced size ($4 \times 4$) by keeping only the four columns corresponding to the nonzero locations in $\mathbf{x}$. In this way, a unique solution can be ensured with only four measurements assuming that $\mathbf{E}_K$ has a full column rank.

column rank, one would only need $K$ ($K \ll N$) measurements (instead of $N$ in traditional acquisition) to ensure a unique solution

$$\mathbf{s} = \mathbf{E}_K \mathbf{x}_K . \tag{8.3}$$

This simple example tells us how sparsity plays a role in reducing the number of measurements in a linear equation. However, it assumes that the locations of the sparse coefficients are known in advance, which is impossible in practice. Thus, more measurements are typically needed in actuality. In general, if we have $2K$ measurements for a $K$-sparse signal (the size of $\mathbf{s} = 2K \times 1$ and the size of $\mathbf{E} = K \times N$, $2K < N$), one can guarantee a unique solution to Eq. (8.2) if every $2K$ columns in the encoding operator $\mathbf{E}$ are linearly independent (see Appendix 8.A for more details) [32]. Here, in addition to signal sparsity, an additional condition is also enforced to the encoding operator $\mathbf{E}$, and this requirement (every $2K$ columns in $\mathbf{E}$ are linearly independent) is essentially what incoherence implies in compressed sensing. This also suggests that the level of sparsity (i.e., the number of nonzero coefficients) determines the minimum measurements required to solve an under-determined linear equation.

With this simple demonstration, we get to know how sparsity plays a role in signal acquisition and the basic understanding of what incoherence is. Now we will discuss how a sparse signal can be recovered in practice and how we can meet the requirement of incoherence.

### 8.2.2 Compressed sensing reconstruction

The aim of compressed sensing reconstruction is to recover a signal from a reduced number of measurements. By definition, the sparsity of a signal $\mathbf{x}$ is calculated by its zero-norm (L0-norm), which is given by counting the number of nonzero coefficients in the signal. As a result, the recovery of a sparse signal in an undersampled linear equation can be formulated as the following L0-minimized optimization problem:

$$\begin{aligned}\tilde{\mathbf{x}} &= \arg\min_{\mathbf{x}} \|\mathbf{x}\|_0 \\ s.t. \quad &\mathbf{s} = \mathbf{E}\mathbf{x}.\end{aligned} \tag{8.4}$$

From Section 8.2.1, we know that Eq. (8.2) has infinite solutions if $M < N$ in the encoding operator $\mathbf{E}$. In this case, Eq. (8.4) aims to find a solution that has the smallest number of nonzero coefficients from all possible solutions. When the requirement of incoherence (as described in 8.2.1) is satisfied, this solution is unique.

In practice, we will face two problems when solving Eq. (8.4). First, few signals are truly sparse, and most signals are not sparse by themselves. For example, the brain MR image shown in Fig. 8.1 is not sparse, but it has a representation in the wavelet domain in which its coefficients decay rapidly with most of its coefficients close to zero, and the wavelet image can be well-approximated as a $K$-sparse signal. In this case, this signal is said to be compressible. We can expand the sensing problem (Eq. (8.2)), so that a compressible signal $\mathbf{x}$ can be expressed with a sparse basis and associated sparse representation as follows:

$$\mathbf{d} = \Phi\mathbf{x} \tag{8.5}$$

Here, $\Phi$ denotes the basis under which $\mathbf{d}$ can sparsely represent the original signal $\mathbf{x}$, and it is assumed to be an orthonormal basis for simplicity. It is also known as a sparsifying transform. With this definition, Eqs. (8.2) and (8.5) can now be combined as

$$\mathbf{s} = \mathbf{E}\mathbf{x} = \mathbf{E}\Phi^{\mathrm{H}}\mathbf{d}, \tag{8.6}$$

where $\mathbf{E}\Phi^{\mathrm{H}}$ combines the encoding operator with the sparsity basis and $\mathbf{H}$ denotes the Hermitian transpose. Second, signal measurements are also contaminated by noise, which needs to be taken into consideration in signal recovery. With these two conditions, Eq. (8.4) can be modified to

$$\begin{aligned}\tilde{\mathbf{d}} &= \arg\min_{\mathbf{d}} \|\mathbf{d}\|_0 \\ s.t. \quad &\left\|\mathbf{s} - \mathbf{E}\Phi^{\mathrm{H}}\mathbf{d}\right\|_2^2 < \varepsilon.\end{aligned} \tag{8.7}$$

Here, the L2-norm is employed to quantify the difference between the measurements and the estimated solution and $\varepsilon$ is related to the noise level in signal measurement.

Challenges to solving Eq. (8.7) are that L0-norm minimization is a computationally intensive problem and real signals are generally compressible but not truly sparse as previously described. Therefore, a relaxed strategy has been proposed to make the optimization problem tractable. Commonly used sparse-recovery algorithms utilize the L1-norm as a surrogate measure of signal sparsity [11,32,33], and the corresponding optimization problem minimizing the L1-norm is given by

$$\begin{aligned} \tilde{\mathbf{d}} &= \arg\min_{\mathbf{d}} \|\mathbf{d}\|_1 \\ s.t. \quad & \left\|\mathbf{s} - \mathbf{E}\Phi^{\mathrm{H}}\mathbf{d}\right\|_2^2 < \varepsilon. \end{aligned} \tag{8.8}$$

Here, the L1-norm is defined as the summation of the absolute values of all coefficients in a signal

$$\|\mathbf{d}\|_1 = \sum_i |d_i| \,. \tag{8.9}$$

L1-norm minimization is an effective alternative because it promotes sparsity, and it is a convex problem guaranteeing a global minimum, as shows in Fig. 8.3. L1-norm minimization has also been shown to have the same solution as L0-norm minimization under appropriate conditions [11,32,33]. Instead of L1-norm minimization, different approaches have also been proposed to solve the compressed sensing problem based on Lp-norm minimization with $0 < p < 1$ [34–36]. This likely leads to a sparse solution, as seen in Fig. 8.3c, but it results in a nonconvex optimization problem and requires more complex algorithms to find the right solution while avoiding local minima.

Alternatively, we can also reformat Eq. (8.9) as

$$\begin{aligned} \tilde{\mathbf{x}} &= \arg\min_{\mathbf{x}} \|\Phi\mathbf{x}\|_1 \\ s.t. \quad & \|\mathbf{s} - \mathbf{E}\mathbf{x}\|_2^2 < \varepsilon. \end{aligned} \tag{8.10}$$

The constrained optimization problem in Eqs. (8.8) and (8.10) can be further formulated as an unconstrained problem using Lagrange multipliers as shown below, and they can be efficiently solved using different optimization algorithms previously described in Chapter 3:

$$\tilde{\mathbf{d}} = \arg\min \left\|\mathbf{s} - \mathbf{E}\Phi^{\mathrm{H}}\mathbf{d}\right\|_2^2 + \lambda \|\mathbf{d}\|_1 \tag{8.11}$$

$$\tilde{\mathbf{x}} = \arg\min \|\mathbf{s} - \mathbf{E}\mathbf{x}\|_2^2 + \lambda \|\Phi\mathbf{x}\|_1 \tag{8.12}$$

Eq. (8.11) is known as the synthesis formulation of compressed sensing reconstruction since it aims to reconstruct the sparse representation of the signal under the basis $\Phi$, and the final signal to be reconstructed needs to be generated (or synthesized) as $\Phi^{\mathrm{H}}\mathbf{d}$. Eq. (8.12) is called the analysis formulation that directly reconstructs the signal $\mathbf{x}$. When the sparsifying transform $\Phi$ is orthonormal, these two formulations are equivalent.

Note that both Eqs. (8.11) and (8.12) are nonlinear, thus requiring a nonlinear optimization algorithm. A regularization parameter $\lambda$ is required to control the balance between the sparsity term (term on the right) and the consistency with acquired measurements (on the left). With L1-norm minimization, more measurements are expected to ensure successful signal recovery. It has been shown that under proper conditions, a $K$-sparse signal can be reconstructed with high probability if the number of measurements $M > \mu^2 K \log^4 N$ [11,33], where $\mu$ indicates the coherence level. In practice, a good rule of thumb is to have ~3–5 $K$ measurements to recover a $K$-sparse signal based on L1-norm minimization [37,38].

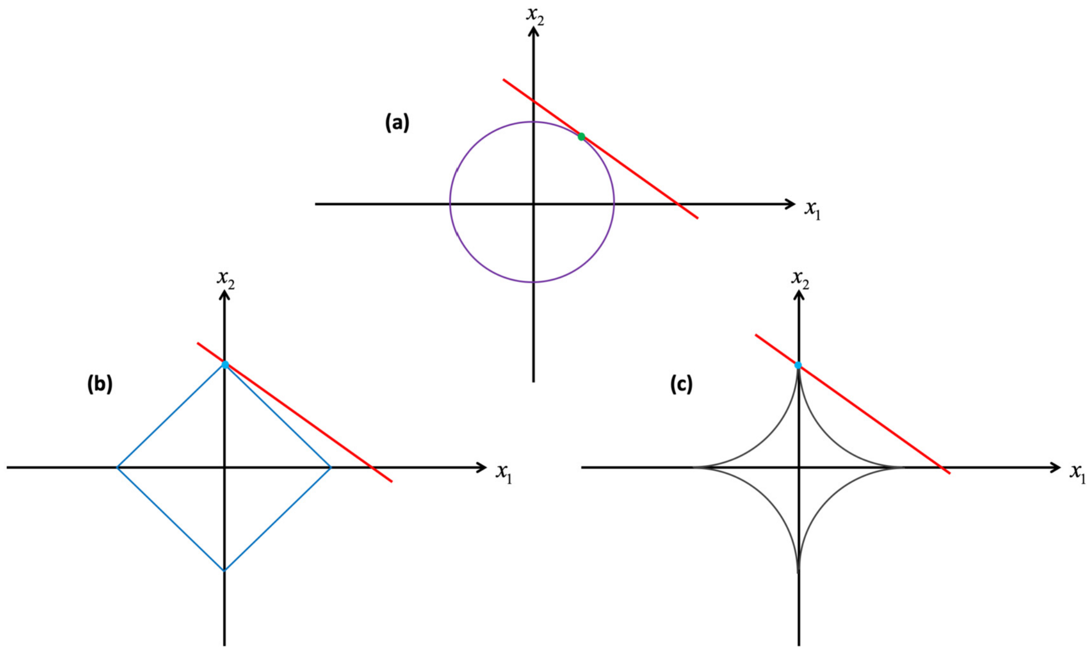



**FIGURE 8.3**

Plots of unit circles for **(a)** the L2-norm (purple), **(b)** the L1-norm (blue) and **(c)** the Lp-norm $(0 < p < 1)$ (black) calculated for a two-element vector $[x_1, x_2]$. The red line represents a linear equation. The plots suggest that (i) a sparse solution (i.e., $x_1 = 0$ in the blue dot) can be obtained based on L1-norm optimization and Lp-norm $(0 < p < 1)$, while the use of L2-norm fail to promote sparsity (green dot); (ii) although both L1-norm and Lp-norm $(0 < p < 1)$ promote sparsity, L1-norm is convex to guarantee a global minimum while Lp-norm $(0 < p < 1)$ is nonconvex (color figure is only available on the web).

### 8.2.3 Conditions for compressed sensing reconstruction

Section 8.2.1 describes the role of sparsity in sparse signal recovery and what incoherent sampling is. Section 8.2.2 describes how a sparse or compressible signal can be recovered. This section presents how to choose a sparsifying transform and how to design an incoherent sampling scheme.

Compressed sensing involves three domains, including: the original signal domain, the sampling domain in which a signal is measured, and the sparse domain in which a signal can have a sparse representation, as shown in Fig. 8.4. The first requirement for successful compressed sensing reconstruction is that a compressible signal should have a sparse representation in the sparse domain and should not be sparse in the sampling domain. In other words, a signal with a sparse representation in the sparse domain must have a dense representation in the sampling domain. This ensures that sparse signal recovery is possible from only a small subset of measurements. As a specific example, considering the worst scenario with $\mathbf{E} = \Phi$, the sampling domain becomes the same as the sparse domain. In this situation, when samples are not acquired, they are permanently lost and cannot be recovered.

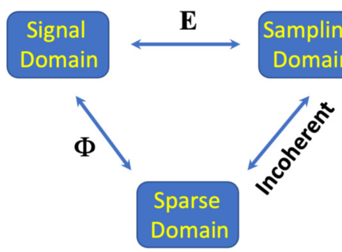



FIGURE 8.4

Compressed sensing reconstruction involves operation in three domains, including the original signal domain, the sampling domain in which a signal is measured, and the sparse domain in which a signal can have a sparse representation. Compressed sensing requires that a compressible signal should have a sparse representation in the sparse domain and should not be sparse in the sampling domain. That means a signal with a sparse representation in the sparse domain must have a dense representation in the sampling domain.

Another component that is important to ensure successful signal recovery in compressed sensing is the design of an undersampling measurement scheme, with which the columns of the encoding matrix can be largely independent in an undersampled linear equation. This is generally referred to as incoherent undersampling in compressed sensing, and the requirement needs to be fulfilled when we design and build the encoding operator $\mathbf{E}$ in Eqs. (8.11) and (8.12). It has been suggested that the mutual coherence of a matrix can be used to easily evaluate the incoherence level of $\mathbf{E}$ [39,40], which is given by the largest absolute inner product between any two columns of $\mathbf{E}$

$$\mu(\mathbf{E}) = \max_{i \neq j} \left| \langle e_i, e_j \rangle \right| \tag{8.13}$$

Eq. (8.13) evaluates the off-diagonal entries of the Gram matrix $\mathbf{E}^{\mathrm{H}}\mathbf{E}$. When the off-diagonal entries of matrix $\mathbf{E}^{\mathrm{H}}\mathbf{E}$ are small (e.g., $\mathbf{E}^{\mathrm{H}}\mathbf{E}$ is close to the identity matrix), $\mathbf{E}$ exhibits low mutual coherence and thus high incoherence. On the other hand, when the off-diagonal entries are large, then $\mathbf{E}$ exhibits high mutual coherence, which is not desired in compressed sensing [39]. The readers will see how this property can be evaluated in compressed sensing MRI in the next section.

## 8.3 Compressed sensing MRI

The compressed sensing theory described in the previous section has three key components, including: 1) sparsity, 2) incoherence, and 3) nonlinear reconstruction promoting sparsity. Accordingly, applying compressed sensing to MRI applications involves three objectives, including: (i) finding the right sparsifying transform to represent an MR image; (ii) constructing a good sensing matrix and sampling pattern for incoherent undersampling (thus acceleration of data acquisition); and (iii) building an iterative nonlinear optimization algorithm to reconstruct the MR image [13,14]. In this section, we will learn how these requirements can be fulfilled in compressed sensing MRI in the context of single-coil acquisition.

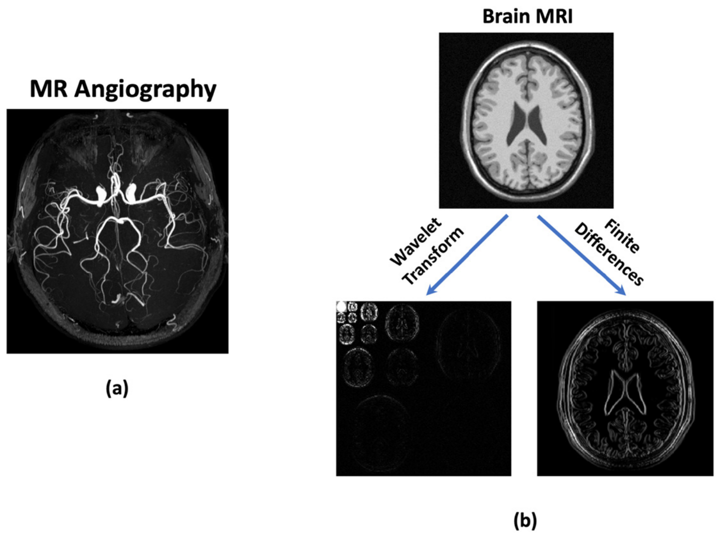



**FIGURE 8.5**

**(a)** An MR angiography image is sparse by itself, with the value of most image coefficients close to zero. **(b)** A brain MR image is not sparse by itself, but it can have a sparse representation after applying a wavelet transform or spatial finite differences.

### 8.3.1 Sparsifying transform and transform sparsity

A signal is sparse if most of its coefficients are zero or approximately sparse if they are close to zero. With this definition, certain MR images (e.g., images from MR angiography) can be considered sparse in the image domain, as shown in Fig. 8.5a. However, most MR images are not sparse, but they typically have a sparse representation in an appropriate transform domain [13]. Popular sparsifying transforms include the wavelet transform and finite differences, as shown in Fig. 8.5b for concrete examples. A combination of two or more sparsifying transforms can also be applied in compressed sensing MRI to improve reconstruction quality [13].

In general, higher image dimensions can result in increased image sparsity. As a result, 3D MRI images can have a sparser representation by employing a 3D sparsifying transform to exploit its pixel correlations in all three spatial dimensions. Dynamic MR images usually have a much higher level of sparsity than static images after applying a temporal sparsifying transform along the dynamic dimension [41], as shown in Fig. 8.6. This is because of the presence of large temporal correlations associated with periodic motion or smooth signal evolution that are often seen in dynamic MRI, similar to temporal correlations in videos leading to higher compression ratio than images. Commonly used generic temporal sparsifying transforms include 1D temporal wavelet transform, temporal Fourier transform, and temporal finite differences. Meanwhile, spatial and temporal sparsifying transforms can be combined to jointly exploit the spatiotemporal correlations in dynamic MR images.

In addition to the generic transforms just mentioned, a sparsifying transform can also be learned from MR images, which could be more effective to obtain a sparser and often more accurate represen-

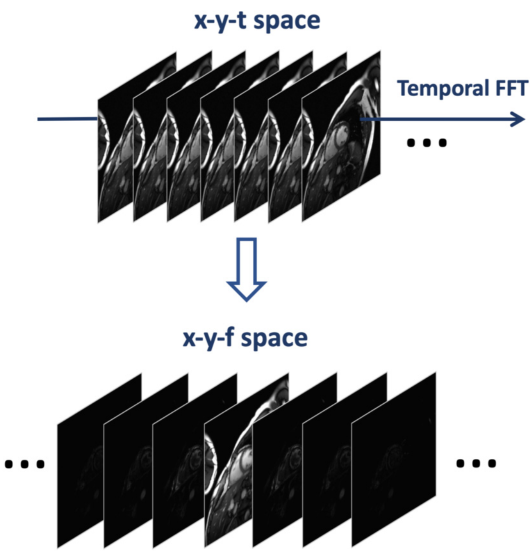



**FIGURE 8.6**

A dynamic image series (x-y-t space) can have a sparse representation by applying a temporal sparsifying transform such as a temporal Fourier transform (x-y-f space). Dynamic MR images typically have a much higher level of sparsity than static images because of the presence of significant temporal correlations associated with periodic motion or smooth signal evolution that are often seen in dynamic MRI.

tation [42–45]. Specifically, a sparse representation can be learned from an MRI database consisting of similar images in advance or from the to-be-reconstructed undersampled images directly [16]. Low-rank-based MRI reconstruction, as will be seen in more detail in the next chapter, is also related to the concept of compressed sensing. They both aim to reconstruct sparse images from data acquired with an incoherent undersampling scheme using a nonlinear iterative reconstruction algorithm.

### 8.3.2 Incoherent data acquisition

In addition to image sparsity, successful implementation of compressed sensing MRI requires the conditions described in Section 8.2.3. For the first condition, since MRI acquisition is performed in the Fourier domain (k-space), this naturally ensures that any single measurement in k-space has contribution to all pixels in the image domain. In other words, an MR image that is sparse in the image domain will have a dense representation in the sampling domain (k-space), and thus it is possible to reconstruct the image, even if some measurements (k-space data) are omitted. Therefore, as long as the selected sparsifying transform basis is sufficiently incoherent with the measurement basis (Fourier basis in this case), this condition will be satisfied.

For the second condition, the nature of sequential k-space sampling in MRI offers the flexibility to acquire a user-defined subset of k-space measurements, in most cases. Here, considering standard 2D Cartesian acquisition with 1D undersampling, $\mathbf{E}$ represents undersampled Fourier basis (size $= M \times N$). According to Eq. (8.13), one can assess whether $\mathbf{E}$ has low mutual coherence by checking the

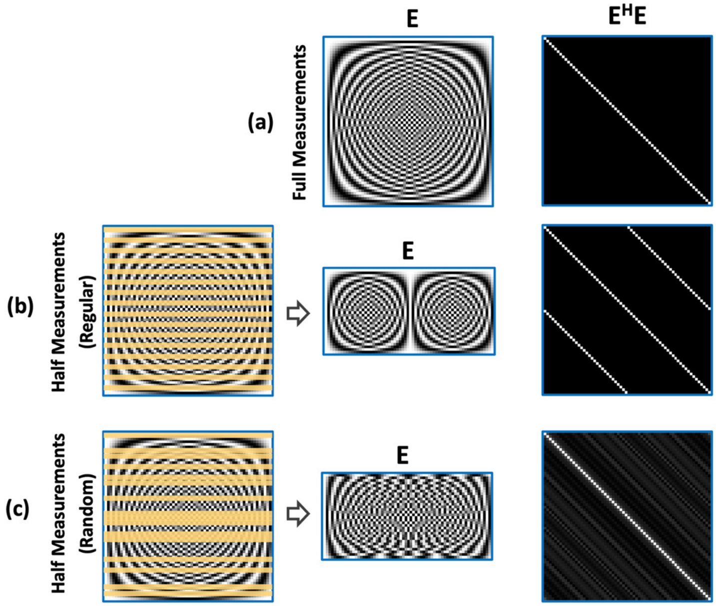



**FIGURE 8.7**

Demonstration of the effect of undersampling scheme with Fourier encoding. For the fully sampling case **(a)**, $\mathbf{E}$ represents the full Fourier basis to implement fast Fourier transform (FFT). The resulting measurement matrix thus has a low mutual coherence, as reflected by the zero off-diagonal entries in the Gram matrix $\mathbf{E}^{\mathbf{H}}\mathbf{E}$. When only half of the Fourier measurements are sampled regularly **(b)**, the measurement matrix has high mutual coherence, as indicated by the replication of diagonal entries in the Gram matrix $\mathbf{E}^{\mathbf{H}}\mathbf{E}$. When half of the k-space lines are sampled randomly, the measurement matrix has low mutual coherence, as indicated by the low off-diagonal entries in the Gram matrix $\mathbf{E}^{\mathbf{H}}\mathbf{E}$.

off-diagonal entries of $\mathbf{E}^{\mathrm{H}}\mathbf{E}$. Several undersampling schemes are given in Fig. 8.7, including: (a) the full Fourier basis (no undersampling); (b) 2-fold undersampled Fourier basis generated by taking every other row from the full Fourier basis; and (c) 2-fold undersampled Fourier basis generated by randomly taking half of the rows from the full Fourier basis. It can be seen that random undersampling generates lower off-diagonal values than regular undersampling. This simple example shows the connection between the compressed sensing theory and practical sampling scheme design in MRI, suggesting random undersampling as a potential scheme for implementation of compressed sensing MRI [13].

In the original demonstration of compressed sensing MRI, Lustig et al. proposed an easier way to evaluate this condition by calculating the point spread function (PSF) as a measure of incoherence, where low sidelobes in the PSF suggest high incoherence, so that undersampling-induced artifacts should look like added noise and most image content can be preserved [13]. The PSF of 2D Cartesian undersampling patterns can be obtained by preforming 2D FFT on corresponding zero-filled sampling masks. As shown in Fig. 8.8, high incoherence can be obtained using a random undersampling pattern resulting in incoherent artifacts, and high coherence is obtained using a regular undersampling pattern resulting in replication of the image. It should also be noted that, in practice, undersampling is typ-

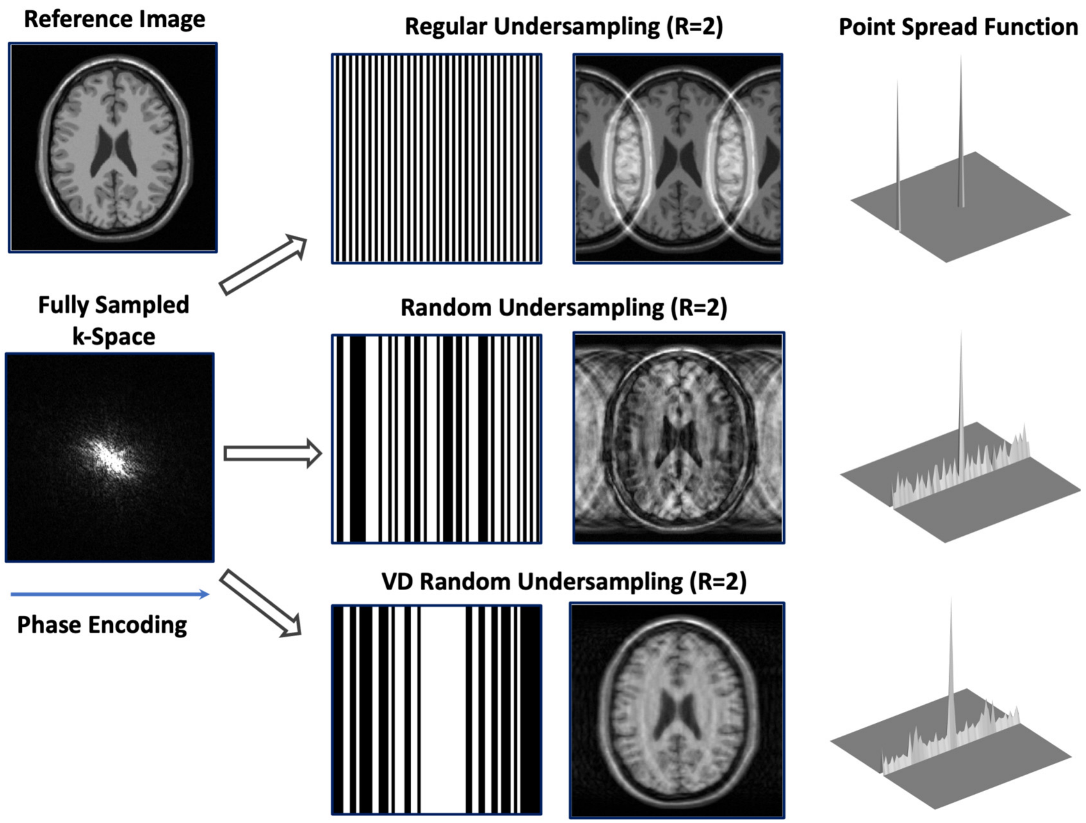



**FIGURE 8.8**

Undersampling in MRI acquisition is typically performed along the phase-encoding dimension. Regular undersampling results in strong coherent aliasing artifacts (top row), while random undersampling results in less coherent artifacts (middle row). Variable-density undersampling, with which the center of k-space is sampled more than the outer region, can further promote incoherence to reduce undersampling-induced artifacts (bottom row). The point spread function (PSF) can be used to evaluate the incoherence of a sampling scheme, where low sidelobes in the PSF indicate high incoherence. The PSF can be obtained by preforming 2D FFT on corresponding zero-filled sampling masks.

ically not implemented along the frequency-encoding dimension in MRI acquisition because it does not save scan time. So, practical implementation of undersampling should take this fact into consideration. Fig. 8.8 also compares undersampled images using a pure random pattern and a variable-density random pattern. The variable-density random pattern samples the k-space center more frequently than the periphery, which results in a cleaner undersampled image. This is because the energy of k-space is mostly located in its center, which is important to determine the image contrast and major contents. Thus, a variable-density undersampling pattern can achieve favorable reconstruction quality compared to a standard uniform undersampling pattern.

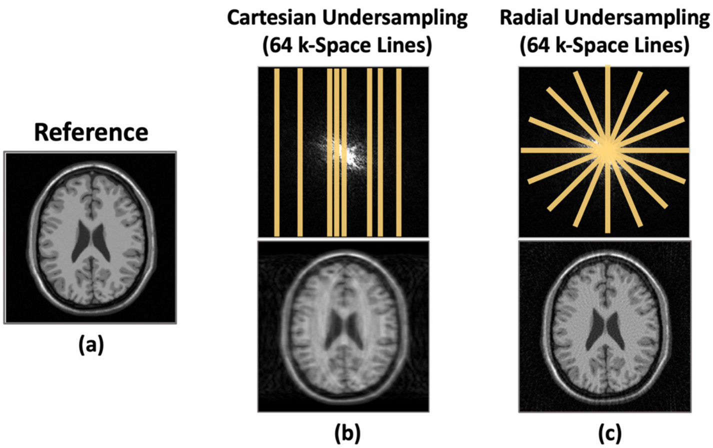



**FIGURE 8.9**

Comparison of undersampled MRI acquisition based on a Cartesian trajectory and a radial trajectory. Although the frequency-encoding dimension is fully sampled for both trajectories, radial sampling does not have a fixed phase-encoding direction, which effectively generates undersampling along two spatial dimensions by omitting certain phase-encoding measurements. As a results, radial undersampling has higher incoherent undersampling behavior than Cartesian undersampling for the same number of measurements.

In addition to random undersampling based on a Cartesian trajectory, various non-Cartesian sampling schemes, such as radial or spiral trajectories, have attracted substantial attention with the introduction of compressed sensing to MRI [15,46–49]. This is because non-Cartesian sampling intrinsically allows for undersampling along all spatial dimensions, thus creating a high level of incoherence. Fig. 8.9 compares undersampling on a Cartesian trajectory and a radial trajectory. Although the frequency-encoding dimension is fully sampled for both cases, radial sampling does not have a fixed phase-encoding direction, which effectively performs undersampling along two spatial dimensions by omitting certain phase-encoding measurements. As non-Cartesian sampling repeatedly acquires k-space center, it also has a lower sensitivity to motion artifacts [50]. However, compared to standard Cartesian MRI, non-Cartesian MRI suffers from several challenges, including increased sensitivity to system imperfections (e.g., off-resonance and gradient-delay effects) and more complex and prolonged reconstruction requiring gridding reconstruction (see Chapter 4) [51]. Selection of desired sampling schemes depends on specific clinical needs, such as reconstruction time, sequence requirements, and motion sensitivity.

3D acquisition generally enables higher acceleration rates compared to 2D acquisition in compressed sensing MRI. This is because: (a) increased sparsity is generally available in 3D images, as previously mentioned; and (b) undersampling in 3D imaging can be performed along two phase-encoding dimensions (including the slice-encoding dimension), which distributes undersampling-induced artifacts along both dimensions to achieve higher incoherence. Fig. 8.10a-b show two different 3D Cartesian random undersampling patterns based on (a) a variable-density random pattern and (b) a Poisson-disc pattern that has more regular random undersampling [21]. Similarly, dynamic MRI can be treated as 3D (2D spatial + 1D temporal) or 4D (3D spatial + 1D temporal) acquisition, making it a

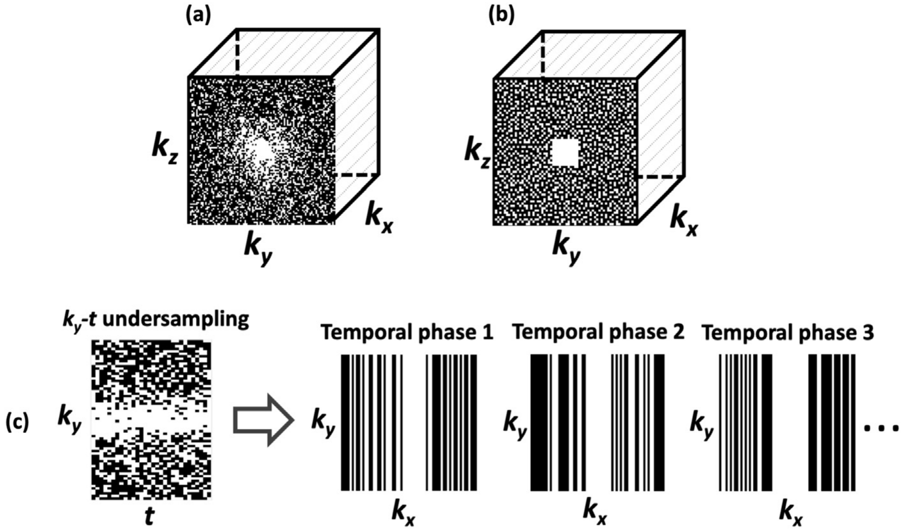



**FIGURE 8.10**

**(a-b)** Two different 3D Cartesian random undersampling patterns based on a variable-density pattern and Poisson-disc distributions, where undersampling can be applied along two phase-encoding dimensions ($k_y$ and $k_z$). For the Poisson-disc pattern, a small region in k-space center is fully sampled to form a variable-density distribution. **(c)** A typical $k_y-t$ (phase encoding and time dimensions) undersampling pattern applied in compressed sensing dynamic MRI, where the undersampling scheme can be varied along time to introduce temporal incoherence. For all the sampling patterns, the frequency-encoding dimension ($k_x$) is fully sampled. Figure is adapted from Feng L et al. (J Magn Reson Imaging 2017 Apr;45(4):966–987 (with permission from John Wiley & Sons, Inc.))

good candidate for the application of compressed sensing methods. For dynamic MRI, high acceleration rates can usually be achieved by varying the undersampling pattern along the temporal dimension. This so-called *ky-t* (phase encoding and time dimensions) undersampling scheme [41], as shown in Fig. 8.10c, is widely used in compressed sensing-based dynamic MRI.

### 8.3.3 Image reconstruction

Most compressed sensing MRI reconstruction methods follow Eqs. (8.11) or (8.12) (for single-coil acquisition). As just mentioned, the reconstruction formulation in Eqs. (8.11) and (8.12) are equivalent if $\Phi$ is orthonormal. However, it should be noted that they can give different solutions when $\Phi$ is a overcomplete basis (e.g., overcomplete wavelets or redundant dictionaries) [52], and it has been shown that the analysis-based model provides more accurate results with overcomplete regularization [53]. Representative algorithms for compressed sensing reconstruction include the iterative soft thresholding algorithm (ISTA) and its different variants, the nonlinear conjugate gradient method, or the alternating direction method of multipliers (ADMM) method. In this section, a simple example is provided

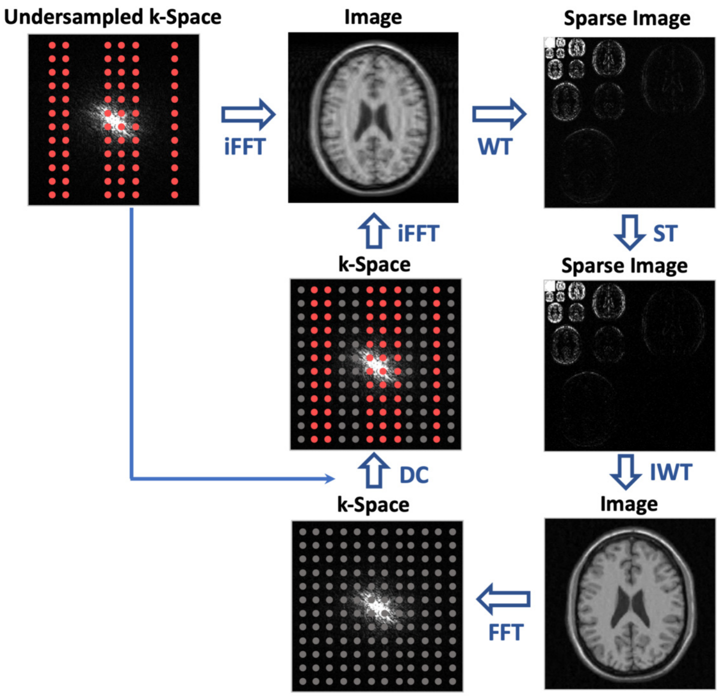



**FIGURE 8.11**

Compressed sensing MRI reconstruction using an iterative soft-thresholding algorithm. Specifically, image reconstruction iteratively loops among three different domains, including the Fourier domain (k-space), image domain, and sparse domain. The reconstruction process implements a soft thresholding process to remove incoherent aliasing artifacts, while maintaining consistency with the acquired undersampled measurements. WT: wavelet transform; IWT: inverse wavelet transform; ST: soft thresholding; DC: data consistency.

to show how we can perform compressed sensing reconstruction to generate an MR image from 2D undersampled Cartesian k-space.

Fig. 8.11 shows a flow chart to demonstrate the reconstruction workflow using ISTA, which can be formulated using the equation

$$\mathbf{d}_{i+1} = \mathrm{ST}_\lambda(\mathbf{d}_i - \Phi^{\mathrm{H}}\mathbf{E}(\mathbf{E}\Phi^{\mathrm{H}}\mathbf{d}_i - \mathbf{s})), \tag{8.14}$$

where the soft thresholding operator is given as

$$\mathrm{ST}_\lambda(\mathbf{d}) = \begin{cases} (|d_i| - \lambda) \cdot \mathrm{sign}(d_i), & \text{if } |d_i| \geq \lambda \\ 0, & \text{if } |d_i| < \lambda \end{cases}. \tag{8.15}$$

This corresponds to the proximal gradient method (see Chapter 3) with a step size of 1. For Cartesian MRI, the practical implementation of compressed sensing reconstruction starts from a zero-filled undersampling k-space, whose size is typically the same as the image to be reconstructed. The zero-filled reconstruction, given by direct Fourier transform of the zero-filled k-space, serves as the starting point of iterative nonlinear reconstruction, as shown in Fig. 8.11. As a result, the encoding operator $\mathbf{E}$ in Eq. (8.12) can be expressed as $\mathbf{BF}$, where $\mathbf{F}$ is the Fourier transform and $\mathbf{B}$ is a binary sampling mask in which 1 indicates locations acquired and 0 indicates locations that are not acquired in the original undersampled k-space.

Here, the reconstruction iteratively loops among three different domains, including: Fourier domain (k-space); image domain (where we generate MR images); and sparse domain (where we have a sparse representation of MR images under a sparsifying transform). Specifically, for each step, a sparsifying transform (a 2D wavelet transform in this case) is performed on the image to obtain its sparse coefficients, followed by a soft thresholding process in the wavelet domain to remove aliasing artifacts, which is equivalent to L1-norm minimization. Here, we can treat the threshold as the regularization parameter shown in Eq. (8.11), and the threshold can be pre-selected or can be adapted in different iteration steps. After the soft thresholding step, the image can be updated as $\mathbf{d}_i - \Phi^{\mathrm{H}}\mathbf{E}(\mathbf{E}\Phi^{\mathrm{H}}\mathbf{d}_i - \mathbf{s})$, where the index $i$ indicates the $i^{th}$ loop. This step is also known as data consistency. This entire process can be repeated for a number of loops until the reconstruction converges. The ISTA algorithm is invoked to reconstruct an image in the sparse domain ($\mathbf{d}$), and the final image is then given as $\Phi^{\mathrm{H}}\mathbf{d}$.

The example shown in Fig. 8.11 is a simple demonstration for readers to understand the process behind iterative compressed sensing MRI. This example also shows that compressed sensing reconstruction can be intuitively understood as a thresholding process with an additional data-consistency constraint. The source code for this reconstruction example is provided in Section 8.8. A summary of various compressed sensing reconstruction demos that are available online is also given at the end of this chapter.

## 8.4 Combination of compressed sensing MRI with parallel imaging

In this section, joint frameworks combining compressed sensing with parallel imaging are described. The rationale behind this synergistic combination is first explained, and representative techniques will then be presented and discussed.

### 8.4.1 Why compressed sensing + parallel imaging?

Compressed sensing MRI can be combined with parallel imaging to form a synergistic framework that enables to fully exploit the coil sensitivity encoding and sparsity jointly for further acceleration of imaging speed [19,21]. Image sparsity and coil sensitivity encoding are complementary sources of information. On one hand, sparsity can serve as a regularizer to prevent noise amplification in parallel imaging; while on the other hand, multicoil acquisition provides additional spatial encoding capabilities to help reduce incoherent aliasing artifacts, thus enabling for higher acceleration rates.

Going back to the compressed sensing theory described in Section 8.2.1, we learned that the minimum requirement for successful recovery of a $K$-sparse signal using compressed sensing is that every $2K$ columns in the encoding/sensing matrix are linearly independent. Now, suppose we have two completely independent sensors to sample the signal, as shown in Fig. 8.12; then, we could then potentially halve the number of samples in each sensor ($K$ measurements from each sensor) while still maintaining

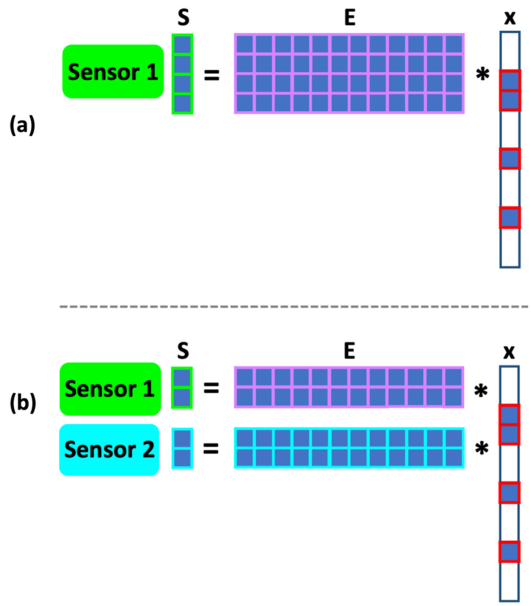



**FIGURE 8.12**

Demonstration of the use of multiple sensors to reduce the number of measurements in compressed sensing. When two sensors are available and if the sensors are completely independent, the number of samples in each sensor could be reduced to half while maintaining the incoherence requirement for sparse signal recovery.

the requirement for sparse signal recovery (a total of $2K$ measurements that are linearly independent). Another way to understand the improved MRI reconstruction quality from a combination of compressed sensing with parallel imaging is that the total number of measurements in the encoding operator ($\mathbf{E}$ in Eq. (8.2)) changes from $M$ to $M * C$, where $C$ indicates the number of coil elements used for MRI acquisition and $M$ is the number of measurements in each element. As a result, we can have $M * C \geq N$ when the number of coil elements is larger than or equal to the acceleration rate, making Eq. (8.2) no longer an underdetermined linear equation and thus more stable.

Translating this multisensor framework to MRI, we can treat multiple sensors as multicoil elements with varying coil-sensitivity profiles. Of course, this simple example does not consider the real situation in MRI that coil elements cannot be fully separated, but it intuitively shows why compressed sensing can benefit from the use of multiple sensors (e.g., multicoil arrays in MRI). Fig. 8.13 provides two specific examples comparing 4-fold and 8-fold undersampled images in both single-coil and eight-coil scenarios, where the multicoil images have a lower level of incoherent artifacts.

### 8.4.2 Representative compressed sensing + parallel imaging methods

There are various ways to combine compressed sensing with parallel imaging. They can be combined in a sequential manner [20] or as a joint reconstruction framework [19,21]. The current consensus is that joint reconstruction is more efficient, and it has been implemented in most of the state-of-art compressed

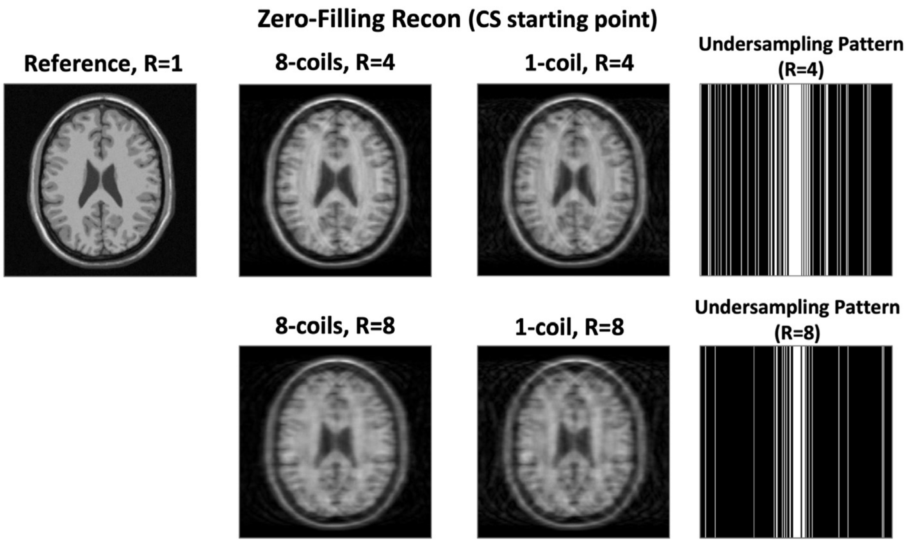



**FIGURE 8.13**

Comparison of 4-fold and 8-fold undersampled images using single-coil and eight-coil acquisitions. The resulting undersampled image from multicoil acquisition shows a lower level of incoherence artifacts compared to single-coil acquisition.

sensing MRI methods. Since parallel imaging can be implemented either in image domain or in k-space, its combination with compressed sensing can also be performed in both domains. Specifically, there are two well-accepted approaches, one combining compressed sensing with parallel imaging using an iterative SENSE-type framework (known as SparseSENSE [19] or its variants) and the other one combining them using a SPIRiT-type framework (known as L1-SPIRiT [21]). Before going to details, readers should refer to Chapter 6 for the principles of both SENSE and SPIRiT.

The SparseSENSE reconstruction framework can be expressed in a same way as in Eq. (8.12). For multicoil Cartesian acquisition, $\mathbf{s}$ becomes multicoil k-space (size $= C * M \times 1$ with $C$ indicating the number of coil elements after concatenating into a vector), $\mathbf{x}$ denotes the image to be reconstructed (size $= N \times 1$), and the encoding operator $\mathbf{E} = \mathbf{BFC}$ combining Fourier transform ($\mathbf{F}$), coil sensitivity encoding with $C$ coil elements (or coil sensitivity maps, indicated as $\mathbf{C}$), and a binary sampling mask ($\mathbf{B}$) as in the single-coil case (see Section 8.3.3). As mentioned, the size of $\mathbf{E}$ is now $M * C \times N$. Compared to single-coil reconstruction, the L1-norm term in SparseSENSE enforces joint multicoil sparsity from coil-combined images. As in standard SENSE reconstruction, the coil sensitivity maps typically need to be pre-estimated from an additional calibration scan. The SparseSENSE framework can be extended to dynamic MRI reconstruction by employing a temporal sparsity transform, in which case the framework is referred as k-t SPARSE-SENSE [22]. This technique is also called L1-regularized SENSE reconstruction in some studies [54] since it uses a L1-norm constraint to suppress noise amplification in iterative SENSE reconstruction. The SparseSENSE framework can also be implemented with non-Cartesian acquisition by replacing the Fourier transform operation with a nonuniform FFT

(NUFFT) operation in the encoding operator **E** [15,46]. Please refer to Chapter 4 for more details about non-Cartesian reconstruction.

The L1-SPIRiT reconstruction can be extended from the SPIRiT reconstruction by adding an additional regularization term promoting sparsity as follows:

$$\tilde{\mathbf{s}}_\mathbf{r} = \arg\min_{\mathbf{s}_\mathbf{r}} \|\mathbf{s} - \mathbf{B}\mathbf{s}_\mathbf{r}\|_2^2 + \lambda_1 \|(\mathbf{G} - \mathbf{I})\mathbf{s}_\mathbf{r}\|_2^2 + \lambda_2 R(\mathbf{F}^H \mathbf{s}_\mathbf{r}). \tag{8.16}$$

Here, $\mathbf{s}_\mathbf{r}$ becomes the to-be-reconstructed multicoil k-space in a Cartesian grid (size $= N * C \times 1$). **B** is a linear operator that relates $\mathbf{s}_\mathbf{r}$ to the acquired undersampled k-space **s**. In the case of Cartesian acquisition, **B** simply selects acquired k-space locations from $\mathbf{s}_\mathbf{r}$, while in the case of non-Cartesian acquisition, **B** becomes an interpolation operator to transfer Cartesian k-space ($\mathbf{s}_\mathbf{r}$) onto a non-Cartesian grid in which data points in **s** are sampled. The differences inside the L2-norm are concatenated as column vectors to enforce data consistency (the left term) and kernel consistency (the middle term—see Chapter 6 for detail), respectively. A regularization can be placed on the image (the right term) to enforce sparsity. After image reconstruction, an image **x** can be generated from $\mathbf{s}_\mathbf{r}$ with a Fourier transform followed by multicoil combination (e.g., sum of square). In the original implementation of L1-SPIRiT, a L1,2-norm was proposed to exploit the joint sparsity of multicoil phased array images as

$$\left\| \begin{matrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_C \end{matrix} \right\|_{1,2} = \sum_{j=1}^{N} \left( \sqrt{\sum_{i=1}^{C} x_{ij}^2)} \right), \tag{8.17}$$

where $C$ and $N$ indicate the number of coil elements and the number of pixels in the image, respectively, and $i$ and $j$ represent the coil and pixel indexes, respectively. The inner L2-norm is used to exploit correlations between different coil elements, and the out L1-norm is employed to enforce sparsity. Alternatively, as demonstrated in the original paper [21], L1-SPIRiT can also be implemented in image domain considering that the SPIRiT operations in k-space are linear shift-invariant convolutions.

The performance of SparseSENSE and L1-SPIRiT is expected to be similar, although coil sensitivities are used in different ways (e.g., explicit estimation of coil sensitivity maps in SparseSENSE and calibration of SPIRiT weights in L1-SPIRiT). The introduction of L1-ESPIRiT (Eigenvector-based SPIRiT) later connects these two types of techniques, which enables a SparseSENSE-type reconstruction formulation that can be implemented with reduced computation complexity compared to L1-SPIRiT. More details of ESPIRiT can be found in Chapter 6.

## 8.5 Clinical applications of compressed sensing MRI

Since the initial introduction of compressed sensing to MRI, there has been an explosive growth of various sparse MRI techniques and clinical applications spanning the body from the head to the foot [17,18,24,55–57]. In particular, the combination of compressed sensing with parallel imaging has enabled highly accelerated data acquisition in various MRI studies, and these improvements can be leveraged to further increase spatial resolution and/or temporal resolution. Meanwhile, various vendors have also developed their own compressed sensing-related fast imaging techniques, which now enable routine clinical applications in everyday MRI exams [58–64].

In principle, compressed sensing can be applied to accelerate all MRI scans, spanning from simple 2D imaging to multidimensional dynamic imaging, and these accelerated MRI scans can help reduce the overall MRI exam times and improve patient throughput. However, the achievable acceleration using compressed sensing depends on many factors, including image sparsity/compressibility (e.g., image content and/or the dimensionality of images), and the sequence used for data acquisition that can lead to different contrast-to-noise (CNR) and single-to-noise (SNR). Meanwhile, injection of a gadolinium-based contrast agent can improve the SNR level, which could result in better reconstruction quality.

In general, moderate acceleration rates can be achieved for the application of compressed sensing in 2D MRI. This is mainly because of relatively limited image correlations in a 2D image and the restriction of undersampling to phase-encoding only. Higher acceleration rates can be achieved when applying compressed sensing to 3D MRI due to higher image compressibility/correlations than 2D and the flexibility of undersampling along the two phase-encoding dimensions to generate increased incoherence. For example, compressed sensing has been applied to highly accelerated 3D MR angiography (MRA) [60,89–99] and 3D late gadolinium enhancement (LGE) imaging [65–71].

In addition to accelerated static MRI, dynamic MRI can greatly benefit from compressed sensing, and highly-accelerated dynamic MRI studies have been demonstrated in various clinical applications. This is due to the extensive spatiotemporal correlations in dynamic MR images and the increased flexibility to design a time-varying undersampling scheme for improving incoherence [72]. For example, many studies have demonstrated highly accelerated cardiac cine MRI [73–80], phase-contrast cine MRI [81–84], and myocardial perfusion MRI [22,85–96]. Some of these techniques have already seen increased routine clinical use [59,79]. Dynamic contrast-enhanced MRI (DCE-MRI) is another application that has been increasingly combined with compressed sensing. The fast imaging speed offered by compressed sensing enables better capture of contrast changes for various oncological applications in the brain, neck, chest, and various body organs [46,97–103]. Compressed sensing has also been combined with non-Cartesian sampling schemes to enable free-breathing DCE-MRI in the chest and abdomen, where it is important to minimize motion-induced artifacts [18]. More application of compressed sensing to high-dimensional MRI application can be found in Chapter 9.

The use of compressed sensing has also been applied to many other MRI techniques and acquisitions on the research level, which could potentially promote the adoption of these techniques for clinical applications. For example, MR parameter mapping, such as T1/T2/R2*/T1rho mapping, can substantially benefit from the use of compressed sensing for accelerated data acquisition [44,104–114]. Other applications of compressed sensing in MRI can be seen in time-resolved MR angiography [115–117], arterial spin labeling (ASL) MRI [117], dynamic musculoskeletal imaging for metal artifact reduction [118], non-proton MRI [119,120], spectroscopic imaging [121–123], and hyperpolarized MRI [124,125]. Due to limited space, these interesting applications cannot be fully covered in this chapter, but readers can refer to the cited references for more details.

## 8.6 Challenges of compressed sensing MRI

As a powerful rapid imaging technique, compressed sensing has achieved a profound impact in MRI since its initial demonstration. However, to date, compressed sensing MRI still needs to deal with several important challenges despite more than a decade of development and optimization. In this final section, we summarize these existing challenges and discuss potential solutions to address them.

The first challenge of compressed sensing MRI is the relatively long reconstruction time compared to non-iterative reconstruction (e.g., parallel MRI reconstruction). Currently, compressed sensing MRI has been implemented by multiple vendors, and it has been broadly adopted clinically for which the reconstruction time does not impact the clinical workflow. However, the slow reconstruction speed still remains a challenge for reconstructing large datasets (e.g., dynamic 4D MRI reconstruction) and non-Cartesian datasets. To overcome this problem, researchers have proposed various off-line reconstruction platforms to bring compressed sensing into the clinic. For example, Block et al. have developed a software tool that can automatically transfer MRI raw data to an external server [126]. Iterative image reconstruction can be performed in the server, and reconstructed images are automatically transferred back to the PACS system. Other similar software tools have also been proposed during recent years [65,127]. However, even with these software tools, radiologists may still experience substantial delays between data acquisition and image generation, making its dissemination challenging. Recent adoption of deep learning for MRI reconstruction has provide a nice solution to address this problem for accelerated 2D or even 3D MRI [28,29,128], as will be more evident in Chapter 11, but it is still challenging to apply such methods to high-dimensional MRI reconstruction due to the need for reference images and memory limitations in training deep neural networks.

Second, compressed sensing MRI requires proper selection of a regularization parameter (or more than one, when more regularizations are used) to control the contribution from the sparsity constraint. As evident in Fig. 8.14, a very large regularization value can lead to overregularization, causing loss of detail, while a too small parameter may result in insufficient suppression of aliasing artifacts. With proper signal normalization and estimation of baseline noise level, the parameter selection may become automatic, and commercially available implementations have extensively demonstrated the robust reconstruction performance of such selection scheme in a large number of clinical scans. However, compared to conventional qualitative MRI, quantitative MRI reconstruction may be more sensitive to the selection of regularization parameters and requires additional care.

Third, compressed sensing has introduced new types of artifacts that are different from artifacts generated from more conventional fast imaging techniques (e.g., parallel imaging). These artifacts include blurring, loss of resolution and contrast, and blocky artifacts. These artifacts could be due to too much compression (excessive acceleration) and/or over regularization. Sometimes the artifacts can be subtle and thus require additional attention from the reading radiologists.

Fourth, since the compressed sensing performance is dependent on the level of sparsity, its acceleration capability is somehow related to the dimensionality of images to be reconstructed. Although compressed sensing still provides moderate advantage in accelerating 2D static compared to parallel imaging, there is a consensus in the field that the achievable acceleration rates using compressed sensing can be higher in 3D MRI and dynamic MRI. However, it should be noted that the computational burden is also increased for high-dimensional MRI reconstruction, posing a trade-off.

Fifth, compressed sensing employs a nonlinear reconstruction process, which changes noise statistics, and the true noise level is hard to estimate. For example, one can simply generate a very clean but completely wrong image by using a very high regularization parameter. As a result, most commonly used metrics for assessing image quality, such as the signal-to-noise ratio (SNR) or contrast-to-noise ratio (CNR), cannot be applied. To date, there is still a lack of reliable metrics for assessment of images reconstructed using compressed sensing. Instead, researchers often perform reader studies, which rely on experienced radiologists to visually evaluate image quality. This, however, is subjective and suffers from observer-dependent variability.

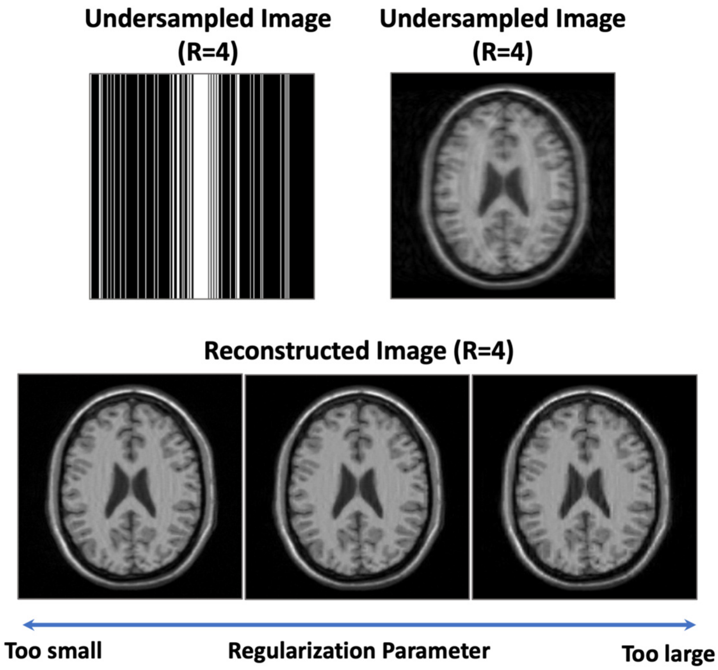



**FIGURE 8.14**

The influence of regularization parameters on compressed sensing MRI reconstruction. A too large value of parameter can lead to over regularization causing loss of details and/or generation of compression artifacts, while a too small value of parameter may result in insufficient suppression of aliasing artifacts.

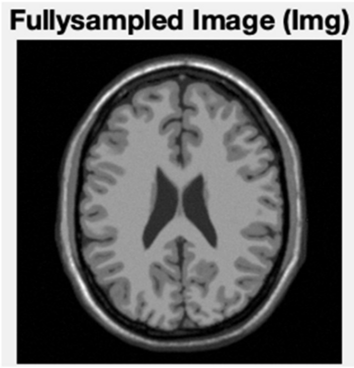
## 8.7 Summary

In this chapter, we learned the three key components of compressed sensing (sparsity, incoherence and nonlinear reconstruction), why we need them, and how they can be implemented in accelerated MRI. Representative frameworks to combine compressed sensing with parallel imaging were presented. Multiple clinical applications of compressed sensing MRI and its existing challenges were summarized. Although image sparsity and incoherent undersampling are usually connected with compressed sensing, it should be realized that both of them play an essential role in more advanced reconstruction techniques, such as the low-rank-based reconstruction, dictionary-based reconstruction, and deep learning-based reconstruction, as will be seen in more details in the following chapters.

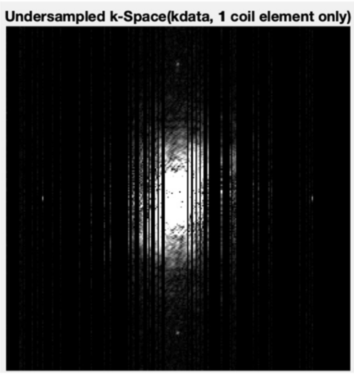
## 8.8 Tutorial

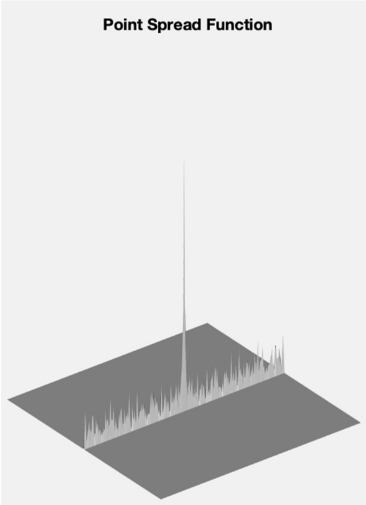
This section provides a demo to reconstruct an undersampled brain image using compressed sensing reconstruction using an iterative soft-thresholding algorithm in MATLAB (Mathworks).

```
clear all;close all;
clc

% Load fully-sampled data
% data: multicoil fully-sampled images
% cmaps: coil sensitivity maps
% mask: 3-fold or 4-fold undersampling pattern
load Brain_8ch.mat;
load('mask_R3.mat') %
cmaps=double(cmaps);
nx,ny,nc=size(data);

% Plot the PSF of the undersampling pattern
PSF=abs(fft2c_mri(mask));
figure,surfl(PSF/max(PSF(:))); colormap(gray),axis square;shading flat,axis
off

% Multicoil combination using sum of square
Img=sum(abs(data.^2),3).^(1/2);
```

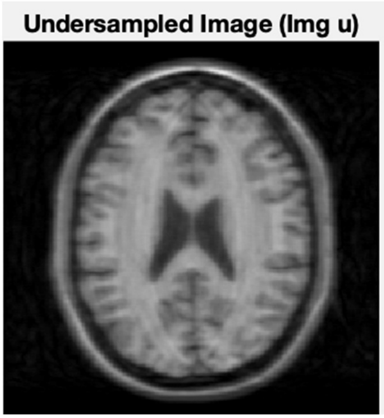



```
% Performing 1D variable-density undersampling
mask=repmat(mask,1 1 8);
kdata=double(fft2c_mri(data).*mask);
```



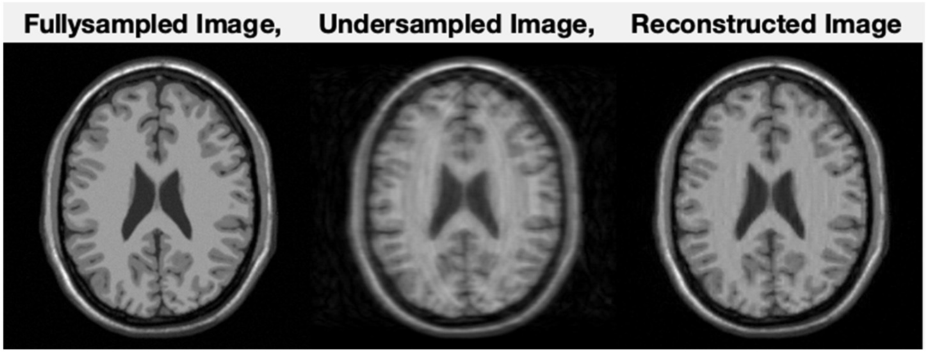



```
% Multicoil Encoding Operator
E=Emat(mask,cmaps);

% Wavelet transform
W=Wavelet('Daubechies',4,4);

% Regularization Parameter
lambda=0.002;

% Number of iterations
nite=50;

% Get undersampled images
Img_u=E'*kdata;
```




```
% From image domain to sparse domain
Img_sparse=W*Img_u;
```

```
% Iterative reconstruction
for ite=1:nite
    Img_tmp=Img_sparse;

    %Soft-thresholding, as shown in Equation 8.16
    Img_sparse=(abs(Img_sparse)
-lambda).*Img_sparse./abs(Img_sparse).*(abs(Img_sparse)>lambda);
    Img_sparse(isnan(Img_sparse))=0;

    % Image update as shown in Equation 8.15 (data consistency)
    Img_sparse=Img_sparse-W*(E'*(E*(W'*Img_sparse)-kdata));

    %Recon information
    fprintf(' ite: %d, update: %f3\n', ite,norm(Img_sparse(:)-
Img_tmp(:))/norm(Img_tmp(:)));
end

%From sparse domain to image domain
Img_recon=W'*Img_sparse;

% Display images (left to right: Fullysampled Image,Undersampled
Image,Reconstructed Image)
figure,imshow(abs(cat(2,Img,Img_u,Img_recon)),0 1),title('Fullysampled
Image,     Undersampled Image,     Reconstructed Image')
```




The source code of this demo, together with the sample dataset, can be found along with this chapter. The readers can try the demo with multiple datasets and undersampling patterns. In addition, there are a number of compressed sensing MRI demos that are available online, such those listed here:

Sparse MRI, L1-SPIRiT, L1-ESPIRiT
http://people.eecs.berkeley.edu/~mlustig/Software.html
SparseSENSE, k-t SPARSE-SENSE
https://cai2r.net/resources/k-t-sparse-sense-matlab-code/
Radial compressed sensing MRI
https://cai2r.net/resources/iterative-reconstruction-from-undersampled-radial-data-using-a-total-variation-constraint-matlab-code/
https://cai2r.net/resources/grasp-matlab-code/

Compressed sensing MRI using the split-Bregman algorithm
https://github.com/HGGM-LIM/Split-Bregman-ST-Total-Variation-MRI
Compressed sensing MRI using the alternating direction method of multipliers (ADMM) algorithm
http://jtamir.github.io/t2shuffling-support/
What's more, various implementation compressed sensing algorithms are available in a software tool called BART (Berkeley Advanced Reconstruction Toolbox), which is also available online.
http://wwwuser.gwdg.de/~muecker1/software.html

## Acknowledgments

The brain MR image used in this chapter was obtained from the ISMRM Sunrise Course on parallel imaging (http://hansenms.github.io/sunrise/sunrise2013/).

## Appendix 8.A Conditions for a unique solution in compressed sensing

This appendix shows why a unique solution can be guaranteed with only $2K$ measurements in an underdetermined linear equation (Eq. (8.2)) if $\mathbf{x}$ is a $K$-sparse signal (at most $K$ nonzero coefficients) and every $2K$ columns in the measurement matrix $\mathbf{E}$ are linearly independent. First, we state the following definitions:

$\mathbb{N} = \left\{\hat{\mathbf{x}} : \mathbf{E}\hat{\mathbf{x}} = 0\right\}$ is the null space of the measurement matrix $\mathbf{E}$;
$\sum K$ is the union of all possible $K$-sparse signals.
$\sum 2K$ is the union of all possible $2K$-sparse signals.

Now, if we have two solutions to Eq. (8.2), $\mathbf{x}_1$ and $\mathbf{x}_2$, and then we have $\mathbf{x}_1 \in \sum K$, $\mathbf{x}_2 \in \sum K$, $\mathbf{x}_1 + \mathbf{x}_2 \in \sum 2K$, $\mathbf{x}_1 - \mathbf{x}_2 \in \sum 2K$, and $\mathbf{E}(\mathbf{x}_1 - \mathbf{x}_2) = \mathbf{E}\mathbf{x}_1 - \mathbf{E}\mathbf{x}_2 = 0$. To guarantee a unique solution, we need to ensure $\mathbf{x}_1 - \mathbf{x}_2 = 0$. This indicates that we will need to have $\mathbb{N} \cap \sum 2K = \{0\}$ so that $\mathbf{x}_1$ must equal to $\mathbf{x}_2$ to satisfy $\mathbf{E}(\mathbf{x}_1 - \mathbf{x}_2) = 0$. This, in turn, suggests that all $2K$ columns in the measurement matrix $\mathbf{E}$ need to be linearly independent to ensure $\mathbb{N} \cap \sum 2K = \{0\}$.

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